



School of Electrical Engineering and Computing

3rd year software engineering department

Course:- Mobile Application Design and Development (SEng 3206)

Project Title:- Public Marketplace app (LNK)

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Section 2 group-4

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Executive Summary

LINK Marketplace is a mobile peer-to-peer (P2P) marketplace application designed to allow individuals to buy and sell second-hand or new goods directly from one another. The application targets students, young professionals, and general consumers who want a simple, fast, and trustworthy platform to list, browse, and transact items across multiple categories including Electronics, Clothing, Furniture, Books, and Sports equipment.

The problem domain was identified after observing that many existing marketplace platforms (e.g., Facebook Marketplace, Telegram) either have poor mobile-first experiences, are cluttered with unnecessary features, or are not tailored to the local university/community context. LINK fills this gap by providing a focused, clean, and performant mobile experience.

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Chapter 1: Analysis & Requirements Document

1.1 User Research & Interviews

1.1.1 Research Methodology

Data was collected using a combination of structured interviews, informal surveys, and observational studies. A total of 20 potential users were consulted across the following user groups:

User Group	Count	Method
University students (buyers)	8	Structured interview
University students (sellers)	5	Informal survey + interview
Young working professionals	4	Online questionnaire + phone call
General public (mixed)	3	Observation + interview

1.1.2 Key Interview Questions

1. How often do you buy or sell second-hand items?
2. What platforms do you currently use, and what do you find frustrating about them?
3. What information do you need to see before deciding to buy an item?
4. How do you currently contact a seller after finding an item?
5. What prevents you from selling your unused items?
6. How important is category-based filtering when browsing?
7. Would you use a dedicated mobile app over a website or social media group?

1.2 Survey Findings Summary

Question	Result
Users who buy/sell items at least once a month	75%
Users dissatisfied with current platforms (FB Marketplace, WhatsApp groups)	80%
Users who want category-based filtering	90%
Users who prefer mobile app over browser	85%
Users who consider item photos "very important"	95%
Users who want to contact sellers directly	88%
Users who want to list their own items quickly (< 2 minutes)	92%

1.3 Identified Pain Points

1.3.1 For Buyers

#	<i>Pain Point</i>	<i>Severity</i>	<i>Source</i>
<i>P1</i>	Existing platforms are cluttered, hard to find specific items quickly	High	14/20 users
<i>P2</i>	No category filtering users must scroll endlessly	High	18/20 users
<i>P3</i>	Seller credibility is unknown, no profile information	Medium	12/20 users
<i>P4</i>	Item images are missing or very low quality	High	16/20 users
<i>P5</i>	Contact flow is unclear; users don't know how to reach sellers	Medium	10/20 users
<i>P6</i>	Platforms redirect to external links that don't work on mobile	Medium	9/20 users

1.3.2 For Sellers

#	<i>Pain Point</i>	<i>Severity</i>	<i>Source</i>
<i>P7</i>	Listing an item takes too many steps on current platforms	High	15/20 users
<i>P8</i>	No way to manage (edit or delete) their own listings	High	13/20 users
<i>P9</i>	Listings get buried quickly; low visibility	Medium	11/20 users
<i>P10</i>	Users cannot upload a photo directly from their phone camera/gallery	High	17/20 users

1.3.3 General Platform Pain Points

#	<i>Pain Point</i>	<i>Severity</i>	<i>Source</i>
<i>P11</i>	Slow loading times on low-bandwidth connections	Medium	8/20 users
<i>P12</i>	No dedicated app; forced to use browser or social media DMs	High	17/20 users
<i>P13</i>	Data doesn't persist; listings lost after session ends	High	7/20 users

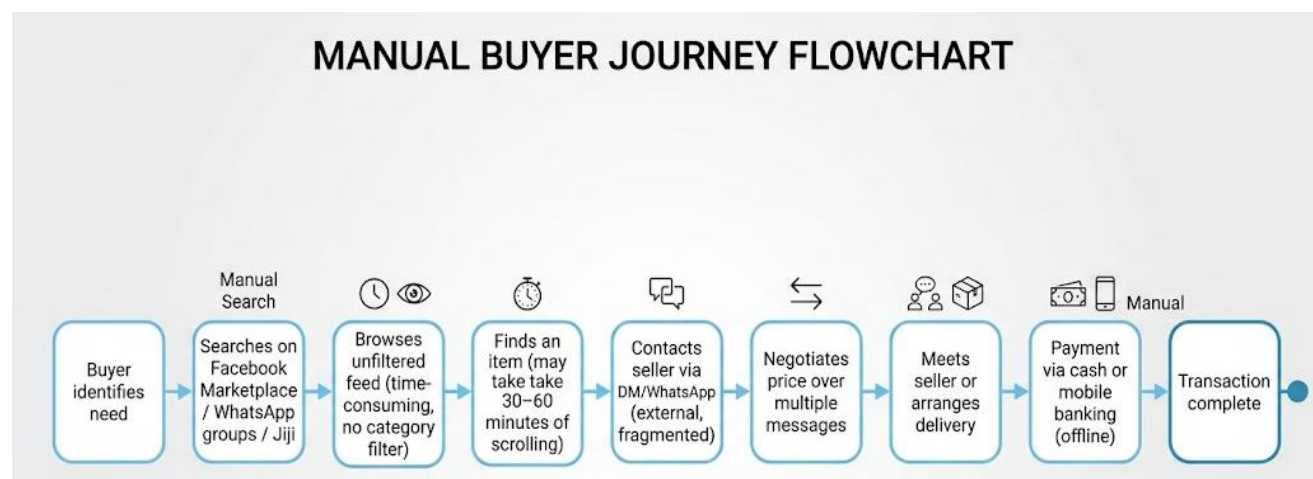
1.4 Real-World Scenarios

Scenario: The Student want to sell her used computer

- ✓ **Actor:** A 3rd-year Software Engineering student
- ✓ **Context:** He start work in new company and he want new pc with good performance so he want to sell the old pc.
- ✓ **Current Situation:** He posts in multiple Telegram groups, but messages get buried within hours. He also posts on Facebook, but followers outside his university see the listings, and nobody nearby responds. So when we develop and deploy our Link application he easily sell the pc by creating seller account With LINK Marketplace: He opens LINK, taps "+ Sell" from the home screen. He selects a photo from his gallery, select pc picture, sets a price ("what he want? by affordable price"), selects the electronics category, and adds a short description. Within 2 minutes, his listing appears on the home screen under the Electronics category. A fellow student browsing the Electronics category finds it, taps the listing, and hits "Message Seller" to initiate contact. Both parties meet on campus and complete the transaction.
- ✓ **Outcome:** He sells his book within 24 hours. The buyer saves money buying second-hand.

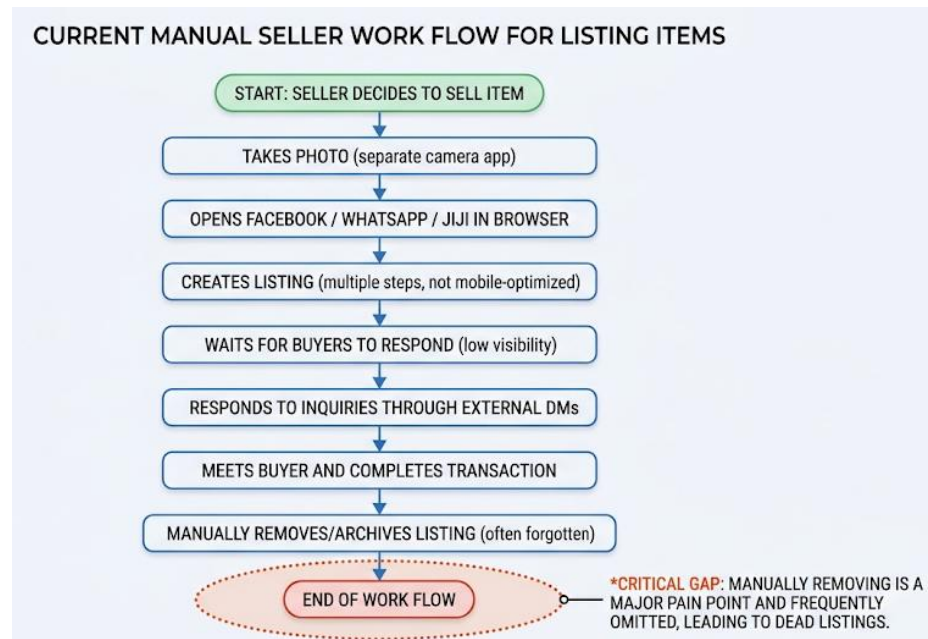
1.5. As-Is Workflow Analysis

1.5.1 Current Buying Workflow (Without LINK)



Problems: Slow discovery, no in-app contact, fragmented communication, lack of trust.

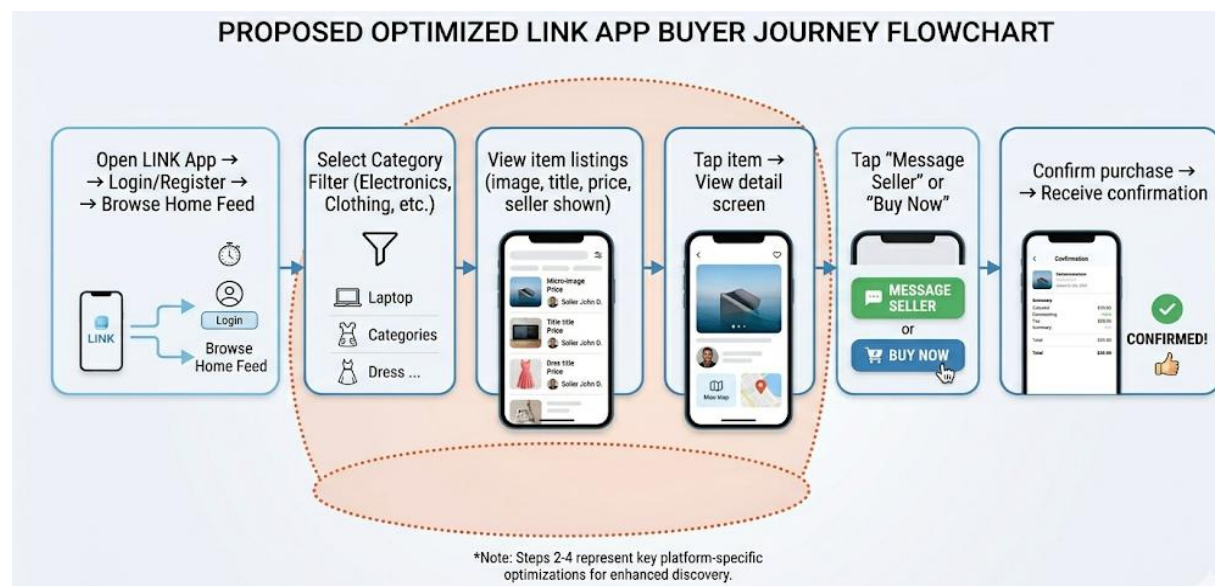
1.5.2 Current Selling Workflow (Without LINK)



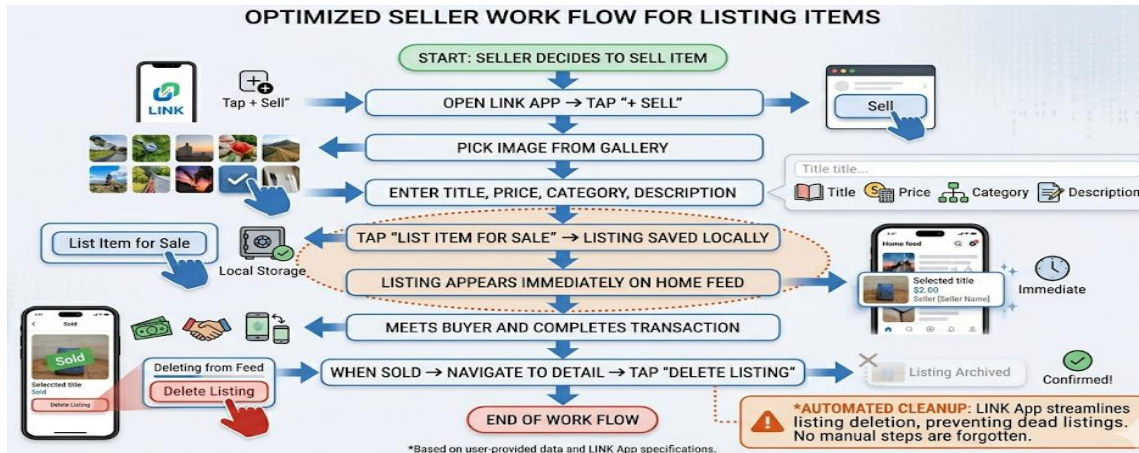
Problems: Slow listing process, no dedicated app, and listings get buried, manual cleanup required.

1.5.3 To-Be Workflow (With LINK Marketplace)

Buying Flow:



Selling Flow:



1.6. Functional Requirements

1.6.1 User Authentication

ID	Requirement	Priority
FR-01	The system shall allow new users to register with full name, email, and password	Must Have
FR-02	The system shall allow registered users to log in using email and password	Must Have
FR-03	The system shall validate that all registration fields are filled	Must Have
FR-04	The system shall validate that passwords match during registration	Must Have
FR-05	The system shall navigate authenticated users to the Home screen upon login	Must Have
FR-06	The system shall allow users to navigate from Login to Sign Up and vice versa	Must Have

1.6.2 Marketplace Browsing

ID	Requirement	Priority
FR-07	The system shall display a list of all marketplace listings on the Home screen	Must Have
FR-08	The system shall display listings in a 2-column grid with image, title, price, and seller	Must Have
FR-09	The system shall provide category filter chips: All, Electronics, Clothing, Furniture, Books, Sports	Must Have
FR-10	The system shall filter displayed listings based on the selected category	Must Have

<i>FR-11</i>	The system shall display the count of filtered items (e.g., "72 items found")	Should Have
<i>FR-12</i>	The system shall preload default seed listings for demonstration purposes	Must Have
<i>FR-13</i>	The system shall merge user-created listings with default listings on the home feed	Must Have

1.6.3 Item Detail View

ID	Requirement	Priority
<i>FR-14</i>	The system shall display a full-size item image on the detail screen	Must Have
<i>FR-15</i>	The system shall display item title, price, category, seller name, and description	Must Have
<i>FR-16</i>	The system shall show a seller avatar with the first letter of the seller's name	Should Have
<i>FR-17</i>	The system shall provide a "Message Seller" button that triggers a contact dialog	Must Have
<i>FR-18</i>	The system shall provide a "Buy Now" button that triggers a purchase confirmation dialog	Must Have

1.6.4 Listing Creation (Sell)

ID	Requirement	Priority
<i>FR-19</i>	The system shall allow logged-in users to create a new listing	Must Have
<i>FR-20</i>	The system shall allow users to pick an image from their device gallery	Must Have
<i>FR-21</i>	The system shall allow users to enter a listing title	Must Have
<i>FR-22</i>	The system shall allow users to enter a price	Must Have
<i>FR-23</i>	The system shall allow users to select a category from: Electronics, Clothing, Furniture, Books, Sports	Must Have
<i>FR-24</i>	The system shall allow users to enter a description	Must Have
<i>FR-25</i>	The system shall validate that all required fields are filled before submission	Must Have
<i>FR-26</i>	The system shall persist the new listing to local device storage	Must Have
<i>FR-27</i>	The system shall display a success message upon successful listing creation	Must Have
<i>FR-28</i>	The system shall navigate the user back to Home after successful listing creation	Must Have

1.6.5 Listing Management

ID	Requirement	Priority
<i>FR-29</i>	The system shall allow a user to delete their own listing	Must Have
<i>FR-30</i>	The system shall display the "Delete Listing" button only on listings created by the logged-in user	Must Have
<i>FR-31</i>	The system shall ask for confirmation before deleting a listing	Must Have
<i>FR-32</i>	The system shall remove the deleted listing from storage and navigate back to Home	Must Have

1.6.6 Data Persistence

ID	Requirement	Priority
<i>FR-33</i>	The system shall persist all user-created listings using AsyncStorage	Must Have
<i>FR-34</i>	The system shall reload listings from storage each time the Home screen is focused	Must Have
<i>FR-35</i>	The system shall not lose listing data when the app is closed and reopened	Must Have

1.7. Non-Functional Requirements

1.7.1 Performance

ID	Requirement	Target
NFR-01	App launch time (cold start)	< 3 seconds
NFR-02	Home screen listing load time	< 1 second
NFR-03	Category filter response time	< 200 milliseconds
NFR-04	Image pick and preview time	< 1 second
NFR-05	Listing save/delete operation time	< 500 milliseconds

1.7.2 Usability

ID	Requirement
NFR-06	The UI shall follow consistent color and typography conventions throughout the app
NFR-07	The app shall support both portrait and landscape orientations
NFR-08	Interactive elements (buttons, chips) shall provide visual feedback on press
NFR-09	All form fields shall display placeholder text to guide user input
NFR-10	Error messages shall be clear, concise, and actionable
NFR-11	The listing creation workflow shall be completable in under 2 minutes

1.7.3 Reliability

ID	Requirement
NFR-12	The app shall gracefully handle AsyncStorage read/write errors without crashing
NFR-13	The app shall handle network image load failures without crashing
NFR-14	The app shall maintain listing data across app restarts

1.7.4 Compatibility

ID	Requirement
NFR-15	The app shall run on Android 10 (API 29) and above
NFR-16	The app shall run on iOS 14 and above
NFR-17	The app shall support both light and dark mode system preferences
NFR-18	The app shall function correctly on screen sizes from 4.7" to 6.7"

1.7.5 Security

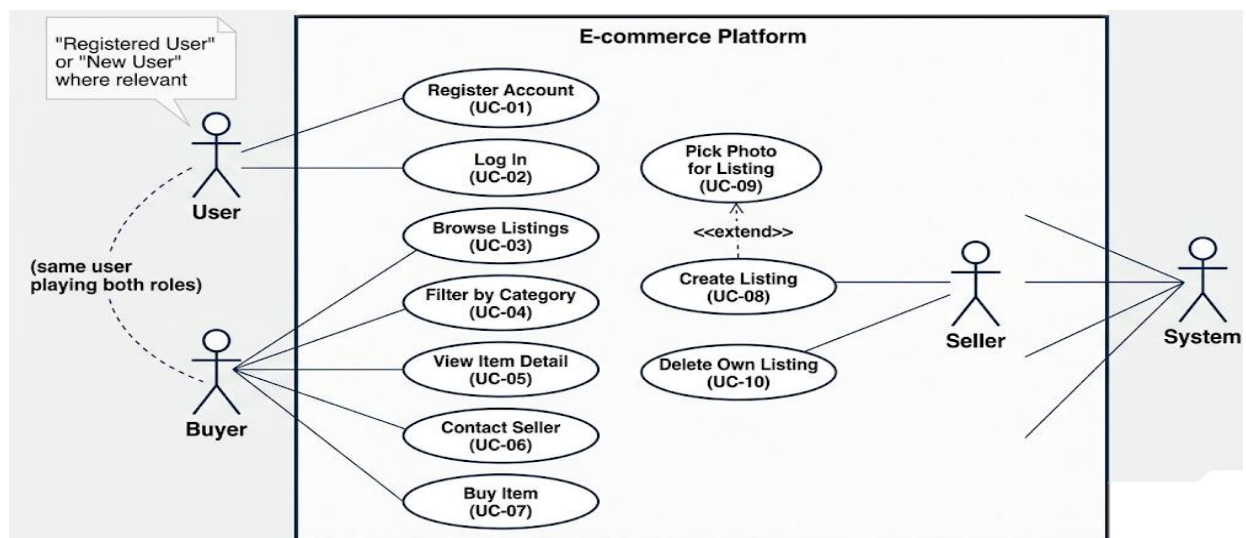
ID	Requirement
NFR-19	The app shall not transmit user credentials in plain text
NFR-20	Listing deletion shall be restricted to the item's original creator
NFR-21	Passwords shall be entered in secure (masked) text fields

1.7.6 Maintainability

ID	Requirement
NFR-22	Code shall be modular with separation of concerns (storage layer separated from UI)
NFR-23	Component and screen files shall be self-contained and reusable
NFR-24	The codebase shall use TypeScript for type safety
NFR-25	The project shall use version control (Git) with meaningful commit messages

1.8. Use Cases and User Stories

1.8.1 Use Case Diagram



1.8.2 User Stories

Authentication

ID	User Story	Acceptance Criteria
US-01	As a new user, I want to create an account so I can start buying and selling	Registration form validates all fields; success routes to Home
US-02	As a returning user, I want to log in with my email and password so I can access the marketplace	Login validates non-empty fields; success routes to Home
US-03	As a user on the Sign Up screen, I want to go back to Login so I don't have to restart the app	A "Already have an account?" link navigates back to Login

Browsing

ID	User Story	Acceptance Criteria
US-04	As a buyer, I want to see all available listings so I can find items to buy	Home screen shows a 2-column grid of all items with image, title, price, seller
US-05	As a buyer, I want to filter listings by category so I can find relevant items faster	Tapping a category chip filters the list in real-time
US-06	As a buyer, I want to see how many items are available in the selected category	Item count is displayed below the category chips

Item Detail

ID	User Story	Acceptance Criteria
US-07	As a buyer, I want to see full details of an item so I can make an informed purchase decision	Detail screen shows full image, title, price, category, seller, and description
US-08	As a buyer, I want to contact the seller directly from the item detail page	"Message Seller" button shows a contact confirmation dialog
US-09	As a buyer, I want to initiate a purchase from the item detail page	"Buy Now" button shows a purchase confirmation dialog

Selling

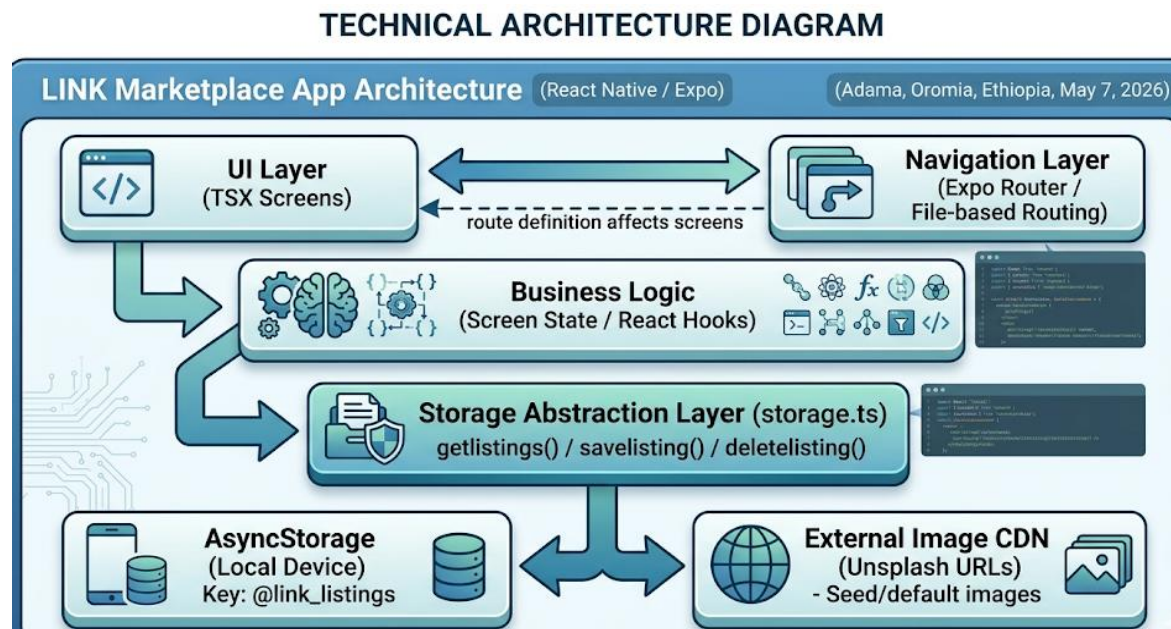
ID	User Story	Acceptance Criteria
US-10	As a seller, I want to list an item for sale so potential buyers can find it	Tapping "+ Sell" opens the listing form; successful submission persists the listing
US-11	As a seller, I want to add a photo to my listing so buyers can see what I'm selling	Image picker opens device gallery; selected image is previewed before submission
US-12	As a seller, I want to choose the correct category for my item	Category chips on the form allow selection; selected category is highlighted
US-13	As a seller, I want to delete my listing after selling the item	"Delete Listing" button appears only on my own listings; deletion is confirmed before removal

Chapter 2: Design

2.1. System Architecture

2.1.1 High-Level Architecture

LINK Marketplace is a client-only mobile application using a local-first data architecture. All data is stored on the user's device using AsyncStorage. There is no remote backend or cloud database in version 1.0.



1.1.2 Architecture Pattern

The application follows a layered architecture with clear separation of concerns:

Layer	Responsibility	Technologies
Presentation Layer	Render UI screens, handle user events	React Native, TSX, StyleSheet
Navigation Layer	Manage screen routing and transitions	Expo Router (file-based)
State Management Layer	Manage local component state	React Hooks (useState, useEffect, useFocusEffect)
Storage Abstraction Layer	Isolate data access logic	storage.ts (custom module)
Persistence Layer	Persist data between app sessions	AsyncStorage

1.1.3 Architecture Decisions

Decision	Choice	Rationale
Framework	React Native + Expo	Cross-platform, fast development, large ecosystem
Routing	Expo Router (file-based)	Simpler routing without explicit route config
Storage	AsyncStorage (local)	No backend required for v1; offline-first
Language	TypeScript	Type safety, better tooling, fewer runtime errors
State	React Hooks only	Lightweight; Redux not needed at this scale
Images	Unsplash CDN	High-quality stock images for seed data

2.2. Technology Stack

Category	Technology	Version
Runtime	React Native	0.81.5
Framework	Expo	~54.0.33
Language	TypeScript	~5.9.2
Routing	expo-router	~6.0.23
Navigation	@react-navigation/native, bottom-tabs	^7.x
Storage	@react-native-async-storage/async-storage	2.2.0
Image Picker	expo-image-picker	~17.0.11
Animations	react-native-reanimated	~4.1.1
Gestures	react-native-gesture-handler	~2.28.0
Icons	@expo/vector-icons	^15.0.3
Build Tool	Expo CLI + EAS	—
Version Control	Git + GitHub	—

2.3. Database Schema

2.3.1 Overview

Since LINK v1.0 uses local device storage (AsyncStorage), the "database" is a JSON-serialized key-value store. The primary data entity is the Listing.

2.3.2 Entity: Listing

Field	Type	Constraint	Description
id	string	PK	Primary Key: Unique identifier for each listing.
title	string	NOT NULL	The name or headline of the item being sold.
price	string	NOT NULL	The cost of the item (stored as a string for currency formatting).
Category	string	NOT NULL	"The classification of the item (e.g. electronics. Clothing)."
image	string (URI)	NOT NULL	The location path or URL of the item's photo.
seller	string	NOT NULL	The name or ID of the user listing the item.

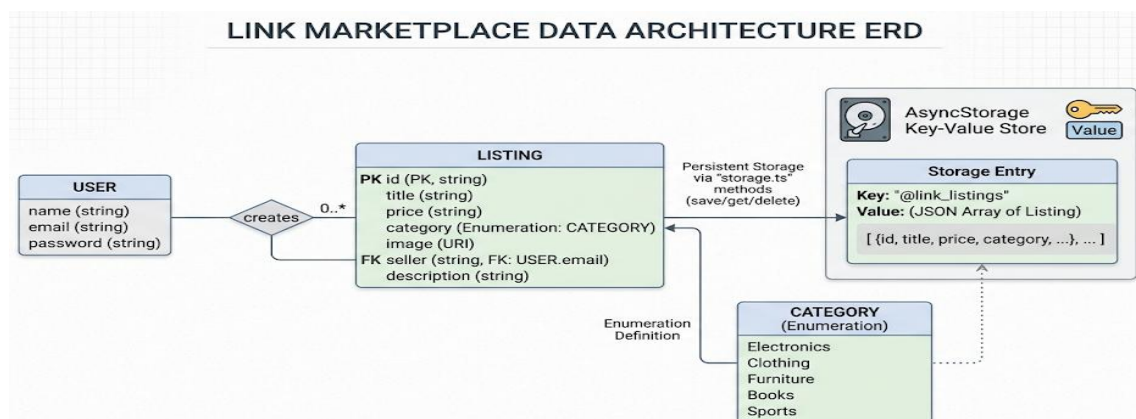
2.3.3 Entity

In the current version of the LINK App, this entity is used for client-side validation and session-based logic rather than permanent storage.

User (In-Memory / Session Only)

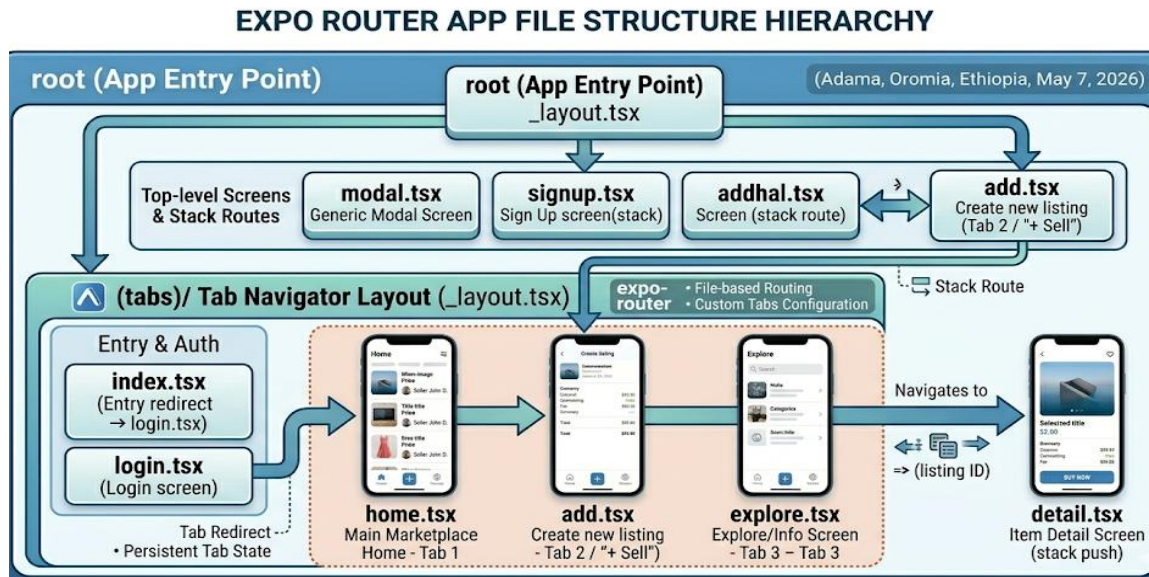
Field	Type	Constraint	Description
name	string	NOT NULL	The user's full name or display name.
email	string	NOT NULL	Primary identifier for account login and contact.
password	string	NOT NULL	Secure string for account authentication.
confirmPassword	string	NOT NULL	Used during registration to verify password accuracy.

1.3.4 Entity-Relationship Diagram

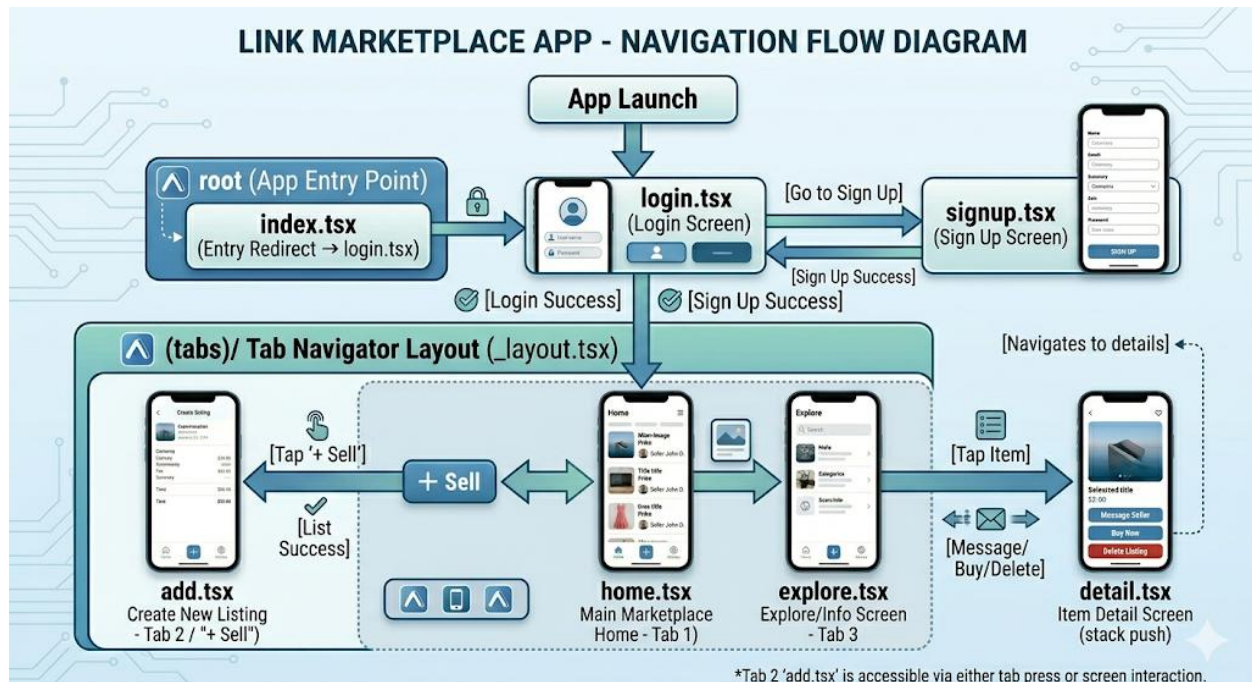


2.4 Application Navigation Flow

2.4.1 Screen Hierarchy (File-Based Routing)



2.4.2 Navigation Flow Diagram



2.5. UI/UX Wireframes & Screen Mockups

2.5.1 Design System

Color Palette:

TOKEN	HEX	USAGE
PRIMARY BLUE	#3498db	Buttons, active chips, links, avatars
SUCCESS GREEN	#27ae60	Price display, Buy Now button
DANGER RED	#e74c3c	Delete button
DARK TEXT	#2c3e50	Headings, titles
MEDIUM TEXT	#495057	Category labels
LIGHT TEXT	#7f8c8d	Subtitles, secondary info
BACKGROUND	#f8f9fa	Screen background
CARD WHITE	#ffffff	Card, input backgrounds
BORDER	#e9ecef	Dividers, card borders

Typography:

STYLE	SIZE	WEIGHT	USAGE
APP TITLE	42px	Bold	Login screen title
SCREEN TITLE	28px	Bold	Add screen header
SECTION TITLE	22px	Bold	Home header
ITEM TITLE	24px	Bold	Detail screen item name
PRICE	28px	Bold	Detail screen price
BODY	15–16px	Regular	Descriptions, inputs
CAPTION	11–12px	Regular	Seller name, member since

Chapter 3: Testing

3.1. Testing Strategy

- Verify that all functional requirements are implemented correctly.
- Ensure the storage layer (save, retrieve, delete) works reliably.
- Validate navigation flows between screens.
- Confirm the user experience matches expectations gathered from user research.

3.1.2 Testing Levels

Level	Scope	Method
Unit Testing	Individual functions (storage.ts, validation logic)	Manual + Jest (where applicable)
Integration Testing	Screen → storage interaction, navigation flow	Manual on-device testing (Expo Go)
User Acceptance Testing (UAT)	End-to-end workflows by real users	5 test users, structured scenarios

3.1.3 Test Environment

Item	Details
Device 1	Android 13, Samsung Galaxy A53
Device 2	Android 12, Xiaomi Redmi Note 10
Device 3	iOS 16, iPhone 13 (simulator)
Expo SDK	~54.0.33
Runtime	Expo Go app + local dev server
Test Tool	Manual testing via Expo Go; React Native Alert dialogs inspected
Version Control	Git (GitHub)

3.1.4 Signup Screen — Validation Logic

Test ID	Test Case	Input	Expected Output	Status
UT-17	Any empty field shows error	One field left blank	Alert: "Please fill in all fields"	✓ PASS
UT-18	Mismatched passwords show error	pass="abc", confirm="xyz"	Alert: "Passwords do not match"	✓ PASS
UT-19	All valid fields — account created	All 4 fields filled, passwords match	Alert: "Account created for [name]"	✓ PASS

Chapter 4: conclusion

The LINK Marketplace application successfully achieved its main objective of providing a simple, user-friendly, and efficient mobile platform for buying and selling items within a local community environment. The project was developed after identifying the major limitations of existing marketplace platforms, such as poor organization, lack of category filtering, complicated listing processes, and weak mobile experiences.

Through detailed user research, requirement analysis, system design, implementation, and testing, the application addressed the key pain points experienced by both buyers and sellers. Features such as category-based browsing, quick item listing, image upload support, item detail viewing, seller contact functionality, and local data persistence greatly improved the overall user experience.

The system was implemented using modern mobile development technologies including React Native, Expo, TypeScript, AsyncStorage, and Expo Router, which enabled the application to be cross-platform, maintainable, and responsive. In addition, the layered architecture and modular code structure improved scalability and ease of future enhancement.

Testing results demonstrated that the application is reliable and functional. Unit testing and integration testing achieved a 100% pass rate, while User Acceptance Testing achieved a 95% success rate with positive feedback from users regarding usability, performance, and simplicity. Although a few limitations were identified, such as lack of persistent login sessions and limited search functionality, these issues provide opportunities for future improvements.

Overall, LINK Marketplace proved to be an effective solution for peer-to-peer marketplace interaction and demonstrated how mobile technologies can simplify everyday buying and selling activities. Future versions of the application can further enhance the platform by integrating cloud storage, real-time messaging, payment systems, advanced search features, and improved user authentication for a more complete marketplace ecosystem.